

UGANDA CHRISTIAN UNIVERSITY

A Center of Excellence in the Heart of Africa

DEPARTMENT OF COMPUTING & TECHNOLOGY

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Question One: PC Hardware and Software Problems, Causes, and Solutions

1. Hardware Problems

Problem A: The Computer Fails to Post (Power On Self-Test) or Turn On

Description: When you press the power button, the PC shows no signs of life whatsoever the fans do not spin, no lights turn on, and the monitor remains blank.

Probable Causes:

- Faulty or Unplugged Power Supply Unit (PSU): The mains power cable might be loose, the switch on the back of the PSU could be turned off, or the internal circuitry of the unit has failed completely.
- Improperly Seated RAM Modules: Dust buildup or physical movement can knock the memory sticks loose, preventing the motherboard from passing its initial startup checks.
- Motherboard Short Circuit or Component Failure: A loose screw, misplaced chassis standoff, or a blown capacitor on the motherboard can short-circuit the board and halt power distribution.

Solutions:

- Check Power Connections and Test the PSU: Ensure the wall socket is switched on and working, check that the PSU switch is set to 'I' (On), and verify that all internal motherboard power cables (the 24-pin and 4/8-pin CPU connectors) are firmly plugged in.
- Reseat the RAM: Turn off the power, unplug the system, remove the RAM modules, clean out the slots with a tin of compressed air, and click the sticks back into place firmly.
- Inspect the Motherboard and Clear the CMOS: Check the board for visible physical damage. Try resetting the BIOS/UEFI settings by taking out the coin-sized CMOS battery for five minutes before popping it back in.

Problem B: Persistent System Overheating and Thermal Throttling

Description: The computer boots up normally but slows down to a crawl or abruptly cuts power and shuts down when you start playing a game or running demanding software.

Probable Causes:

- Dust and Fluff Build-up: A thick layer of dust is blocking the cooling fins of the CPU heatsink and clogging up the case vents, trapping heat inside.
- Dried-out Thermal Paste: The thermal interface paste between the processor and the heatsink has degraded over time, losing its ability to transfer heat effectively.
- Cooling Fan Failure: The CPU fan or case extractor fans have stopped spinning due to mechanical wear or a loose power header on the motherboard.

Solutions:

- Give the System a Deep Clean: Use a tin of compressed air and an anti-static brush to clear out all the dust from the heatsinks, fans, and mesh filters.
- Reapply High-Quality Thermal Paste: Take off the CPU cooler, clean away the old, crusty paste using a bit of high-purity Isopropyl alcohol, and apply a fresh, pea-sized drop of thermal compound before putting the cooler back on.
- Replace or Fixed Broken Fans: Double-check that the fan cables are plugged into the correct motherboard headers. If a fan feels stiff to turn by hand, it is time to replace it.

2. Software Problems

Problem A: Operating System Blue Screen of Death (BSOD) / System Crash

Description: The Windows operating system suddenly stops working, throws up a bright blue screen with an error code (such as DRIVER_IRQL_NOT_LESS_OR_EQUAL), and forces a restart.

Probable Causes:

- Incompatible or Corrupted Device Drivers: A recently updated or corrupted hardware driver (often the graphics or network card driver) is clashing with the Windows kernel.
- Corrupted System Files: Essential operating system files have been damaged due to an unexpected power cut, malware, or an improper shutdown.
- Malware or Virus Infection: A malicious program has integrated itself into core system processes, making the environment completely unstable.

Solutions:

- Boot into Safe Mode and Roll Back Drivers: Fire up the PC in Safe Mode, head over to Device Manager, and uninstall or roll back whichever driver triggered the crash.
- Run a System File Checker (SFC) Scan: Open up the Command Prompt as an administrator and type in sfc /scannow to let Windows automatically find and patch any broken system files.
- Run a Full Antivirus Scan: Use Windows Defender or a reputable third-party antivirus tool to perform an offline boot scan to catch and remove deeply embedded malware.

Problem B: Sluggish Performance and 100% Disk or CPU Utilisation

Description: Applications take an age to open, typing response lags behind, and Task Manager shows that your processor or hard drive is pinned at 100% capacity.

Probable Causes:

- Too Many Startup Programs: A huge queue of background apps is configured to launch the moment you turn on the PC, hogging your memory (RAM) and CPU cycles.
- Severe Disk Fragmentation (On Mechanical HDDs): Data is scattered randomly across a mechanical hard drive's platters, forcing the drive head to work twice as hard to read files.
- Heavy Background Windows Updates or Indexing: Windows Update or search indexing routines are running silently in the background, consuming all available system resources.

Solutions:

- Tweak Your Startup Apps: Open up Task Manager, jump over to the "Startup" tab, and disable non-essential programs (like game launchers and cloud sync utilities) from opening automatically.
- Defragment the Hard Drive (or Optimise SSD): Run the built-in "Defragment and Optimise Drives" utility to tidy up the scattered data on a traditional hard drive.
- Pause or Reschedule Heavy Background Tasks: Go into Windows settings to pause updates or schedule them for a time when you are not using the PC. Let the search index finish its job whilst the computer is idle.

Question Two: Comparison of the OSI and TCP/IP Networking Models

1. The OSI (Open Systems Interconnection) Model

The OSI model is a conceptual framework designed by the International Organisation for Standardisation (ISO) to map out how different computer systems communicate. It is a strictly theoretical blueprint made up of seven distinct layers, engineered to ensure

that hardware and software from entirely different manufacturers can work together seamlessly.

Layer 7: Application Layer: This is where the network interacts directly with software applications (e.g., web browsers using HTTP, email clients using SMTP, or file transfers via FTP).

Layer 6: Presentation Layer: Acts as the translator of the network. It handles data formatting, encryption, compression, and translation (e.g., turning raw data into ASCII, JPEG, or managing SSL/TLS encryption).

Layer 5: Session Layer: Responsible for opening, managing, maintaining, and closing communication sessions between two applications on different devices.

Layer 4: Transport Layer: Ensures reliable, end-to-end data delivery, managing flow control and error checking (e.g., TCP and UDP protocols).

Layer 3: Network Layer: Handles logical addressing and routing, determining the best physical path for data packets to travel across different networks (primarily via IP and ICMP).

Layer 2: Data Link Layer: Manages physical node-to-node data transfer, error detection on the physical layer, and physical addressing using MAC addresses.

Layer 1: Physical Layer: The actual hardware layer that transmits raw, unstructured binary data (bits) over physical mediums like copper cables, fibre-optic lines, or radio waves.

2. The TCP/IP Model (Internet Protocol Suite)

Developed by the US Department of Defense (DoD), the TCP/IP model is a practical, real-world architecture that serves as the actual backbone of the modern Internet. Rather than separating everything out like the OSI model, it condenses network communication into a streamlined four-layer or five-layer framework.

- Application Layer: This layer rolls up the duties of the OSI's top three layers (Application, Presentation, and Session). It talks directly to network applications and includes day-to-day protocols like HTTP, DNS, SSH, and FTP.
- Transport Layer: This corresponds directly to the OSI Transport layer, taking charge of host-to-host communication. It uses TCP for reliable, connection-oriented transfers, or UDP when speed is a priority over reliability.
- Internet Layer: Equivalent to the OSI Network layer. It defines how data packets are structured and routed across networks, rely heavily on the Internet Protocol (IPv4/IPv6).
- Network Access Layer (or Link/Physical Layer): This merges the OSI's Data Link and Physical layers into one. It details exactly how data is physically sent across the local network medium, covering hardware standards like Ethernet and Wi-Fi.

3. Key Differences and Comparison

Feature	OSI Model	TCP/IP Model
Origin	Designed by the ISO (a formal standardisation body).	Developed by the US DoD (a practical, hands-on project).
Number of Layers	7 Layers.	4 or 5 Layers.
Approach	Theoretical; used mostly for teaching and reference.	Practical; the actual blueprint used to run the Internet.
Layer Structure	Explicitly separates the	Merges the top three layers

	Application, Presentation, and Session layers.	into a single Application layer.
Protocol Order	The model was mapped out before the protocols were invented.	The protocols came first, and the model was built around them.
Design Style	Strict, vertical design with very rigid boundaries between layers.	A looser, more flexible framework built to solve practical issues.